

Section Report 3 — Independent Amplicon Analysis

BIOL 4810

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Section Report 3 — Independent Amplicon Analysis

Goal: Use your **own sequencing dataset** to complete the full workflow in Galaxy: process amplicon reads with **DADA2**, then visualize your results in **phyloseq** using **alpha diversity**, **beta diversity**, and a **taxonomic bar plot**.
Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

0) Your dataset

0A. Sample set / project name:

bird microbiome

0B. Sequencing type:

16s v4 paired-end illumina

0C. Metadata grouping variable you analyzed:

sex and population

Workflow overview

For this section report, you will complete:

```
plotQualityProfile → filterAndTrim → learnErrors → dada → mergePairs →  
makeSequenceTable → removeBimeraDenovo → assignTaxonomy → phyloseq alpha  
diversity → phyloseq beta diversity (NMDS using Bray-Curtis) → phyloseq  
taxonomic bar plot
```

Reminder: This section report asks you to apply the entire workflow to your own data and explain the biological meaning of each major output.

Part 1 — Quality profiles

1A) Forward read quality profile

What is the purpose of this step?

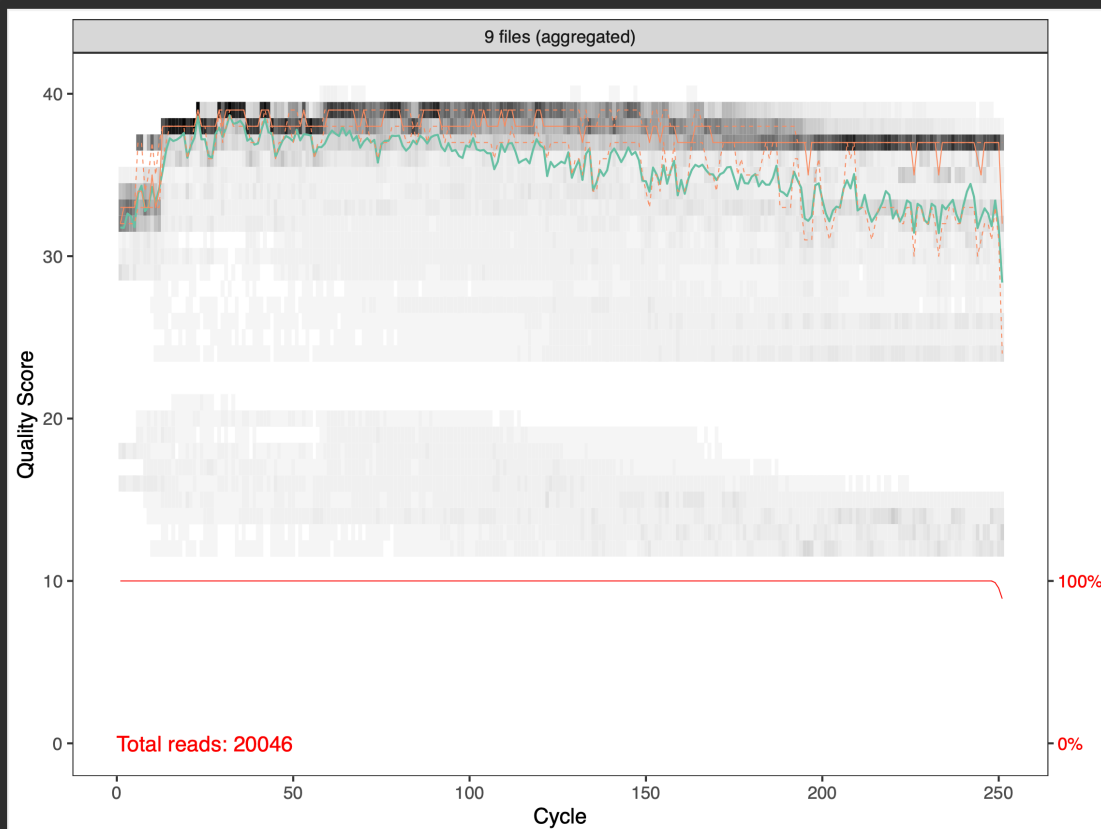
The purpose of this step is to read the profile so that we can filter and trim them. This ensures the quality of the reads.

Upload screenshot: forward plotQualityProfile PDF

Choose File



Screensho....44.51 AM



Where does quality begin to noticeably decline in the forward reads?

The graph doesn't seem to noticeably decline at all.

Chosen forward truncation length:

No truncation

1B) Reverse read quality profile

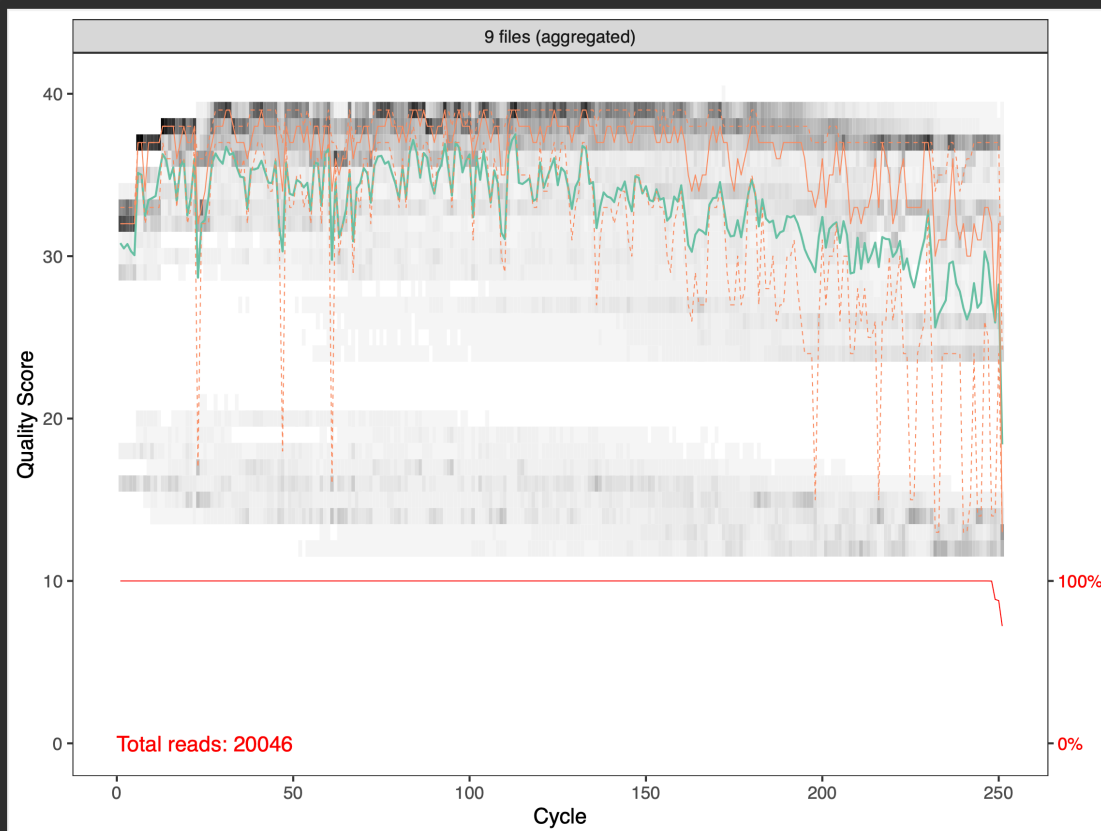
What is usually different about reverse-read quality compared with forward-read quality?

The reverse reads are usually lower quality. You would need to truncate more of the plot.

Upload screenshot: reverse plotQualityProfile PDF

Choose File

Screensho....46.39 AM



Where does quality begin to noticeably decline in the reverse reads?

around 190 bp

Chosen reverse truncation length:

190

1C) Truncation decision

Explain why you chose these truncation values.

I kept my whole forward quality profile because it doesn't seem to have any lower quality peaks. The reverse plot is always lower quality. I truncated it at 190 bp. I didn't want to cut much off because I have a lot of my forward reads.

Rule of thumb: trim where quality begins to drop, but still keep enough overlap for forward and reverse reads to merge.

Part 2 — filterAndTrim

What filterAndTrim does:

This step removes low-quality reads and trims poor-quality bases from the ends of reads before denoising.

Why is this step important?

Removes low-quality reads and trims poor-quality bases from the ends of sequences.

Parameters used:

Separate filters and trimmers for reverse reads- yes
Truncate reverse at 190

Upload screenshot: filterAndTrim output

Choose File

Screensho...7 PM.png

The screenshot shows the Galaxy web interface. The main panel displays the output of the 'filterAndTrim' tool. The table has 2 lines and 3 columns: Column 1, Column 2, and Column 3. The data shows reads.in and reads.out counts. The right panel shows a history of datasets, including 'dada2: filterAndTrim on collection 38: Statistics'.

Column 1	Column 2	Column 3
	reads.in	reads.out
371	3435	3435

History: SR3

dada2: filterAndTrim on collection 38: Statistics

a list with 9 tabular datasets

Download

1: 371

2 lines 3 columns

format tabular, database ?

reads.in reads.out

371 3435 3435

2: 363

2 lines 3 columns

format tabular, database ?

reads.in reads.out

363 1328 1328

3: 352

4: 268

5: 238

6: 216

7: 203

What could happen if you trim too aggressively?

There could be some data loss. There is a possibility that we lose too many base pairs. This makes our results incorrect.

What could happen if you do not trim enough?

This can result in false positives. It will give you data that is untrue.

IMPORTANT — Split Paired Reads (Unzip Collection)

After `filterAndTrim`, your output is a **paired collection**.

Before continuing, you must split this into **forward and reverse collections**.

What to do in Galaxy

1. In the Tools panel, search: **unzip collection**
2. Input: your `filterAndTrim` paired output
3. Run

What you will get

- One collection = **forward reads (R1)**
- One collection = **reverse reads (R2)**

Why this matters

`learnErrors` and `dada` must be run **separately** on forward and reverse reads.
If you skip this, `mergePairs` will fail.

Part 3 — learnErrors

What learnErrors does:

This step models the sequencing error rates in your dataset so DADA2 can distinguish real sequence variants from sequencing mistakes.

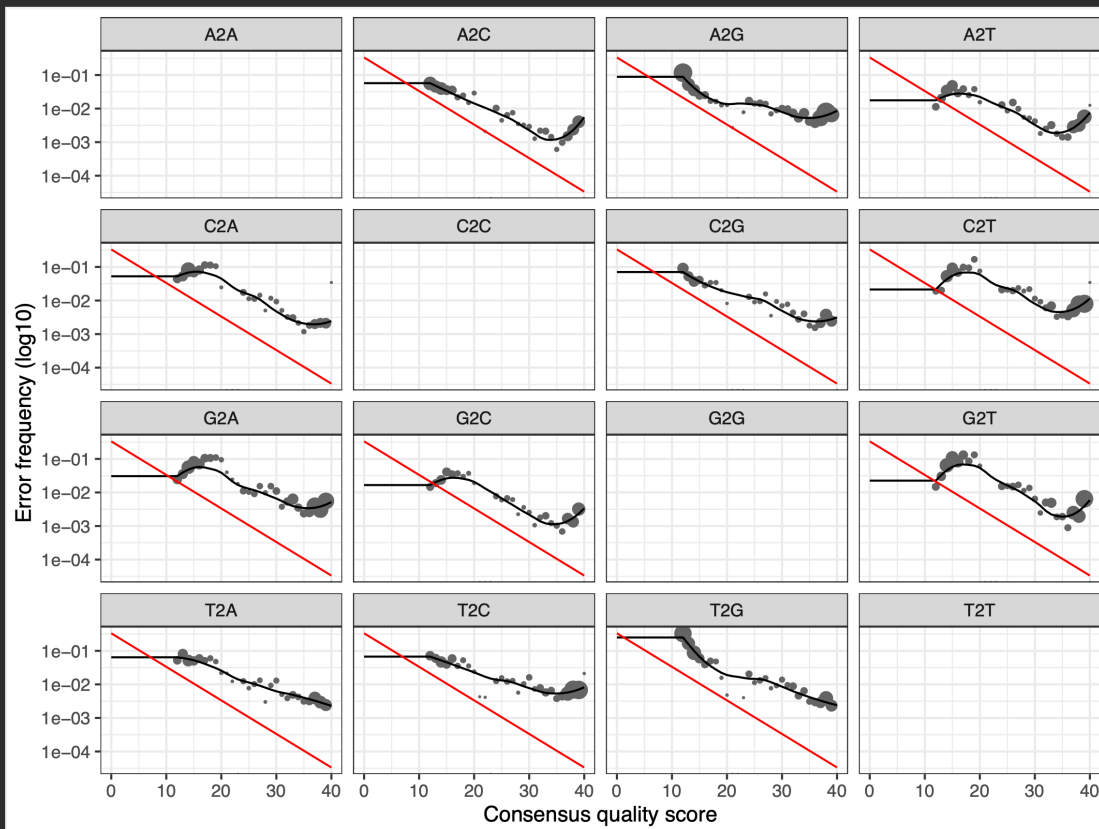
Why is this step important?

This step removes and corrects the data so we don't misidentify the species.

Upload screenshot: learnErrors PDF

Choose File

Screensho....23.55 PM



Does the fitted model generally follow the observed error pattern?

mostly yes

Why is learning error rates important before denoising?

Learning error rates is important before denoising because it removes any noise without destroying any important data.

Part 4 — dada (denoising)

What dada does:

This is the denoising step. DADA2 uses the learned error model to correct likely sequencing errors and infer exact amplicon sequence variants (ASVs).

What is denoising doing?

It removes errors and separates to amplicon data sequences.

How is ASV inference different from OTU clustering?

ASV inference differentiates the sequences by only a single nucleotide. OTU Clustering merges multiple taxa into one taxonomic unit.

Part 5 — mergePairs

What mergePairs does:

This step merges the forward and reverse reads for each amplicon so you can reconstruct the full sequence from overlapping read pairs.

What is being merged, and why?

The filtered forward and reverse reads are being merged to create a single more accurate sequence.

Why do forward and reverse reads need to overlap?

The reads need to overlap because it allows for comparison of the two reads and choose the base with the better quality.

If reads fail to merge, what is one likely reason?

The amplicon is longer than the forward and reverse reads combined.

Part 6 — makeSequenceTable

What makeSequenceTable does:

This step creates the ASV table. Each row is a sample, each column is an ASV, and each cell contains the number of times that ASV was observed in that sample.

What does this table represent biologically?

The table shows the abundance and distribution of the different taxa across the multiple samples.

What are the rows, columns, and cell values in the sequence table?

Rows are the unique DNA sequence. The columns are the individual sample. The cell values represent the count of the sequence within a specific sample.

Upload screenshot: sequence table output

Choose File SequenceT...tPut.png

Dashboard

Galaxy

Section Report 03

Galaxy Europe

Using 19% of 250.0 GB eperez

272: dada2: makeSequenceTable on dataset 263-271 and collection 262

619 lines 10 columns

format dada2_sequencetable database ? size 165.2 KB

Preview

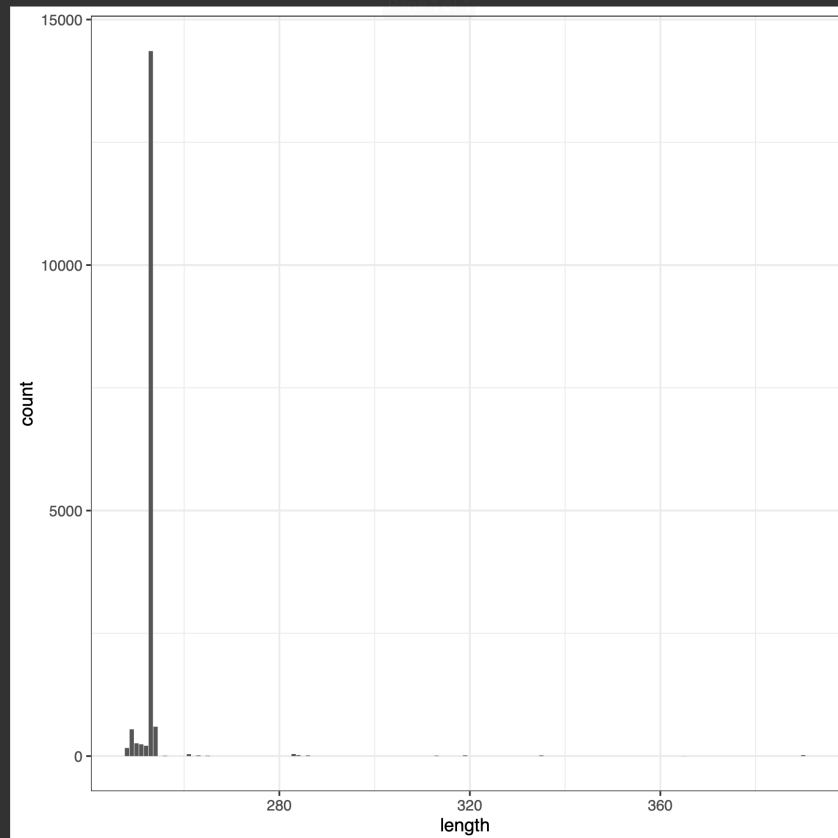
Visualize

Details

Edit

	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10
CCAAAGGCGAAAGCACTCTGCTGGGCCAACTAGCACTGAGAGACGAAAGCTAGGGGAGCAAATGGG	371	363	352	268	238	216	203	97	95
:ACCAAGTGGCGAAGGCGGCTTTCTGGTCTGTAAGTACGCTGAGGCTCGAAAGCGTGGGGAGCAAACAGG	788	0	0	0	0	0	0	0	0
:ACCAACGGCGAAGCACTCTGCTGGGCCGACACTGACACTGAGAGACGAAAGCTAGGGGAGCAAATGGG	0	0	392	42	0	0	0	0	0
:ACCAACGGCGAAGGCACTCTGCTGGGCCGACACTGACACTGAGAGACGAAAGCTAGGGGAGCAAATGGG	0	0	0	0	0	0	0	0	423
:ACCAACGGCGAAGGCACTCTGCTGGGCCGACACTGACACTGAGAGACGAAAGCTAGGGGAGCAAATGGG	0	0	224	0	0	15	0	25	0
:ACCAACGGCGAAGGCACTCTGCTGGGCCGACACTGACACTGAGAGACGAAAGCTAGGGGAGCAAATGGG	0	0	0	0	0	0	0	0	216
:ACCAACGGCGAAGGCACTCTGCTGGGCCGACACTGACACTGAGAGACGAAAGCTAGGGGAGCAAATGGG	0	0	0	0	0	0	55	159	0
:CAGTGGCGAAGGCGGCTGACTGGACATGTATTGACGCTGAGGTGCGAAAGCGTGGGGAGCAAACAGG	0	0	15	52	0	0	21	78	8
:ACCGATGGCGAAGGCGAGCCCTGGGTCAAGATTGACGCTCATGCACGAAAGCGTGGGGAGCAAACAGG	84	0	21	0	0	0	59	0	0
:CCGACGGCGAAGGCACTCTGCTGGGCCGAAACTGACACTGAGAGACGAAAGCTAGGGGAGCAAAGGG	137	0	0	0	0	0	0	0	25
:ACCAAGTGGCGAAGGCGACTTTCTGGTCTGTAAGTACGCTGATGTGCGAAAGCGTGGGGATCAAACAGG	0	40	63	0	0	0	26	0	19
:ACCAAGTGGCGAAGGCGGCTTTCTGGTCTGTAAGTACGCTGAGGCTCGAAAGCGTGGGGAGCAAACAGG	0	0	134	0	0	0	0	0	0
:ATCCATGGCGAAGGCGAGCTACCTGGACCAACACTGACACTGAGGCACGAAAGCGTGGGGAGCAAACAGG	0	0	0	0	125	0	0	0	0
:CACCAGTGGCGAAGGCGAGCCCTGGGTCAAGACTGACGCTCATGCACGAAAGCGTGGGGAGCAAACAGG	116	0	0	0	0	0	0	0	0
:ACCGGTGGCGAAGGCGGCGACCTGGTTCATACTGACGCTGAGGCGCGAAAGCGTGGGGAGCAAACAGG	0	0	0	0	0	0	0	115	0
:ACCGGTGGCGAAGGCGGCTCACTGGTCCGGAAGTACGCTGAGGCGACAGCGTGGGGAGCAAACAGG	0	0	0	0	0	0	75	35	0
:ACCAAGTGGCGAAGGCGGCTTTCTGGTCTGTAAGTACGCTGAGGCTCGAAAGTGTGGGGAGCAAACAGG	0	0	109	0	0	0	0	0	0
:ACCGGTGGCGAAGGCGGCCAACTGACCATTAAGTACGCTGAGGCGCGAAAGCGTGGGGAGCAAACAGG	0	0	0	0	0	0	24	85	0
:CAGTGGCGAAGGCGGCTCACTGGACTGATTTGACGCTGAGGTGCGAAAGCGTGGGGAGCAAACAGG	0	0	0	0	0	0	0	108	0

Upload screenshot: sequence length distribution



What does the sequence length distribution show?

The x-axis shows the length of the DNA sequence, and the y-axis shows the count of those sequences. Most of the sequences are approximately 250 to 260 base pairs long. The peak reaches a count around 15,000.

Part 7 — removeBimeraDenovo

What removeBimeraDenovo does:

This step removes chimeras, which are artificial sequences created during PCR when fragments from different templates are incorrectly joined together.

What is a chimera, and why should it be removed?

A chimera is a DNA molecule formed during PCR amplification. It should be removed because it improves accuracy and cleans the data.

Why are chimeras a problem for interpretation?

It can be incorrectly identified as unique taxa, leading to an inaccurate analysis.

How many chimeric reads were removed? How do you know?

1 sequence was removed. It went from having 618 lines to 617 lines.

Part 8 — assignTaxonomy

What assignTaxonomy does:

This step compares your ASVs to a reference database and assigns taxonomy, such as phylum, class, order, family, genus, or sometimes species.

What does taxonomy assignment add to your ASV table?

Each ASV is assigned a taxonomic name. It makes you go from having an unknown species to a known species.

Why might some ASVs not be assigned all the way to species?

There might be some missing information in the data that the database cannot pick up to identify the species.

Part 9 — Alpha diversity

9A) Alpha diversity

In your own words, what is alpha diversity?

Alpha diversity describes how diverse a community is.

9B) Observed richness

What does Observed measure?

Observed measures the richness of a community. The total number of species.

9C) Shannon diversity

What does Shannon measure?

Shannon measures the richness and evenness.

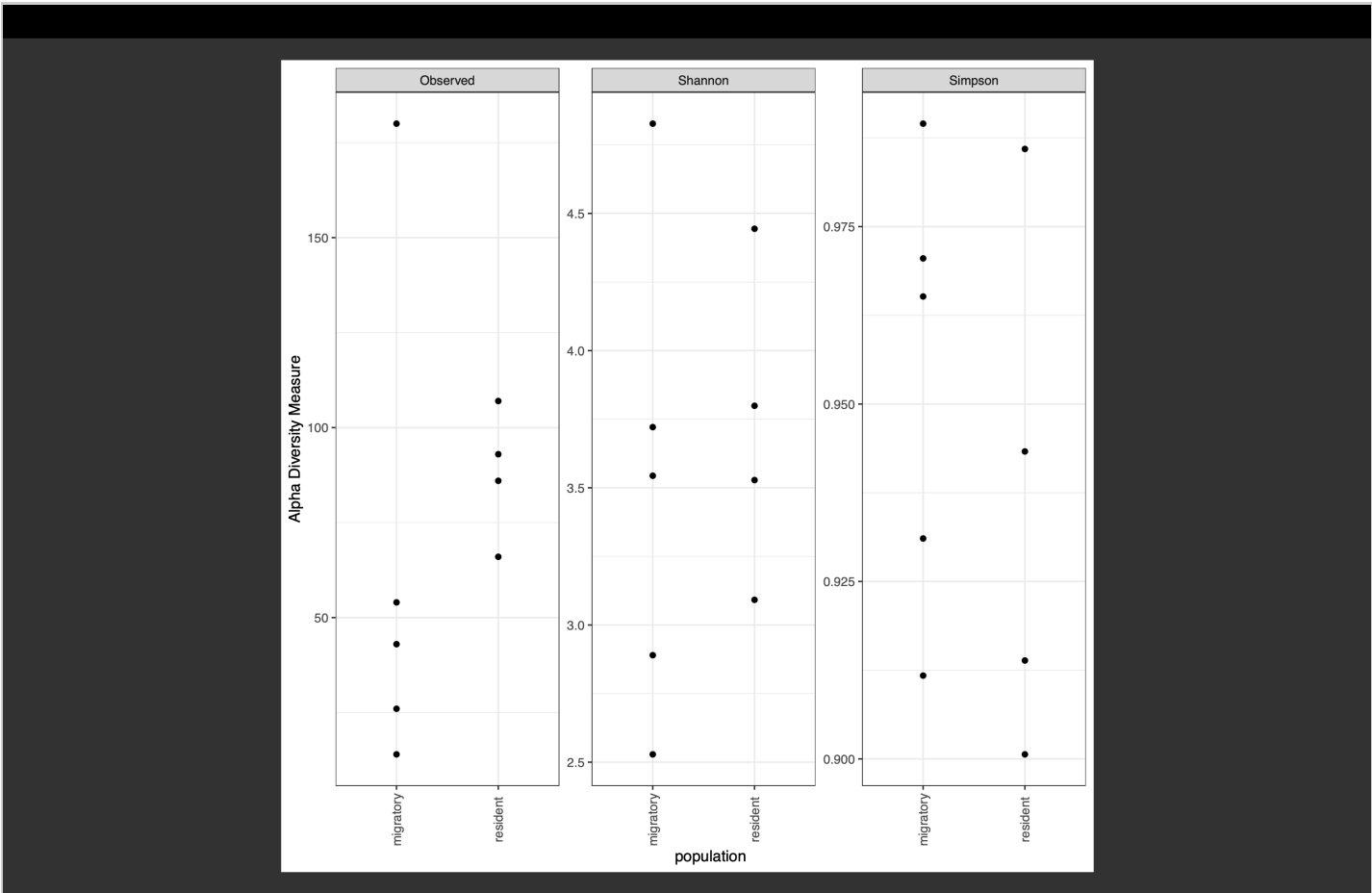
9D) Simpson diversity

What does Simpson measure?

Simpson measures richness and evenness as well, but it is less sensitive to rare species.

Upload screenshot: alpha diversity plot grouped by population

Choose File  Screensho....42.22 PM



Describe differences in alpha diversity by population.

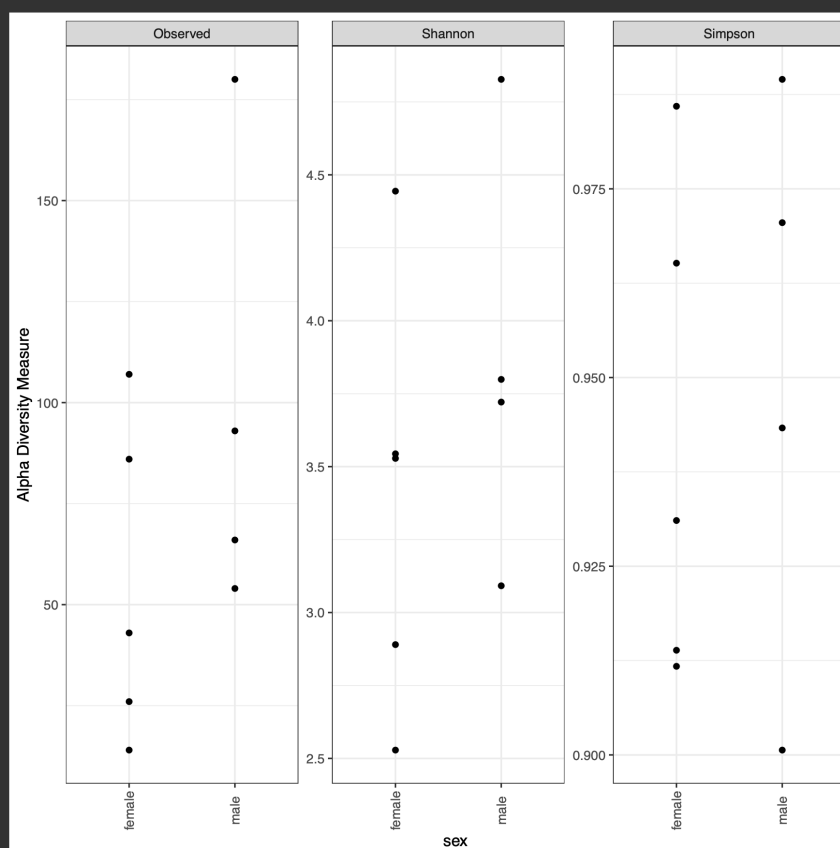
Resident populations have higher observed richness compared to migratory populations. Shannon and Simpson diversity show an overlap between the two groups.

Upload screenshot: alpha diversity plot grouped by sex

Choose File



Screensho....42.49 PM



Describe differences in alpha diversity by sex.

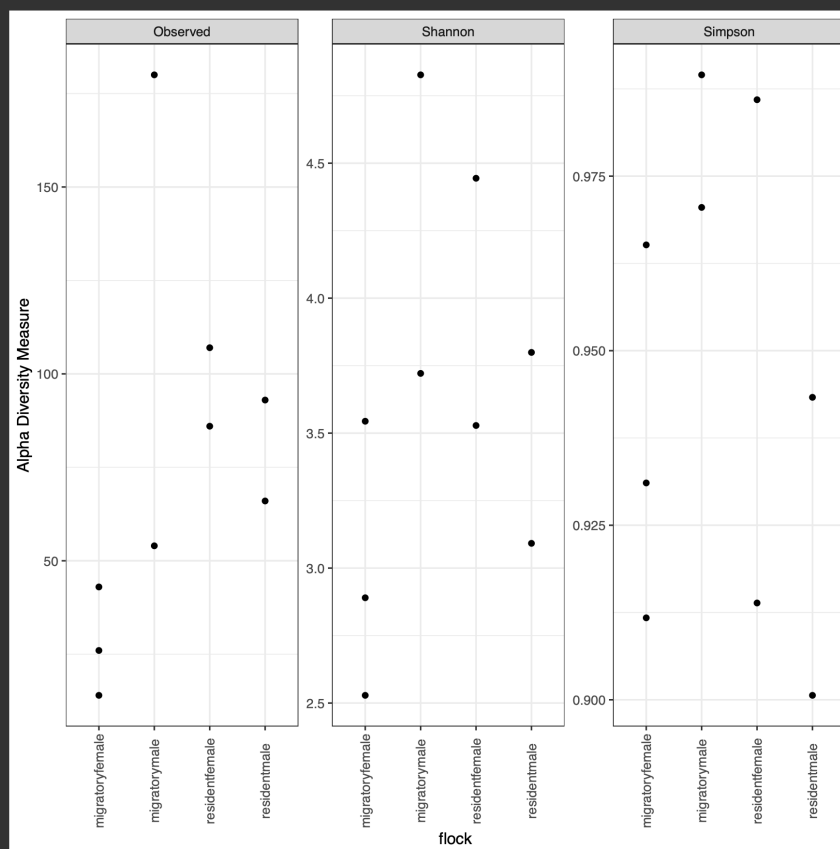
Observed richness is slightly higher in males than in females. You can also see this in Shannon, where the males have a broader range of diversity values, compared to females.

Upload screenshot: alpha diversity plot grouped by flock

Choose File



Screensho....43.14 PM



Describe differences in alpha diversity by flock.

The migratory male flock has the highest observed richness and Shannon diversity. The migratory female flock consistently has the lowest diversity across all three.

Part 10 — Beta diversity (NMDS with Bray-Curtis)

What beta diversity means:

Beta diversity describes differences in community composition **between samples**.

In your own words, what is beta diversity?

Beta diversity measures how similar and different communities are. It compares different communities with each other.

10A) Bray-Curtis dissimilarity

What does Bray-Curtis measure?

Bray-Curtis measures the differences in the microbiome between samples.

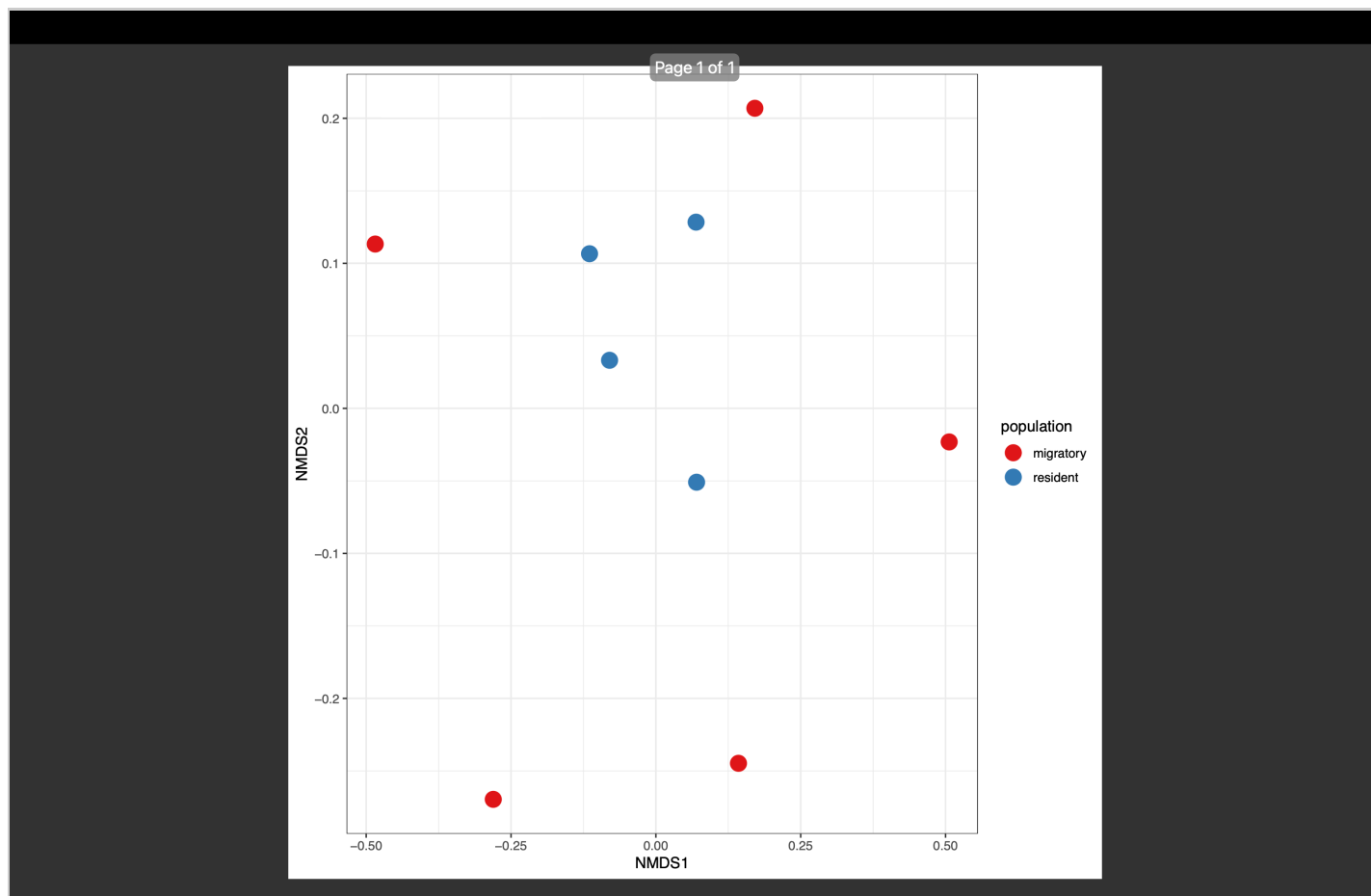
10B) NMDS

What does NMDS show?

NMDS shows the distance between how similar or different the microbiomes are.

Upload screenshot: NMDS Bray-Curtis plot colored by population

Choose File Screensho....46.15 PM



Describe beta diversity patterns by population.

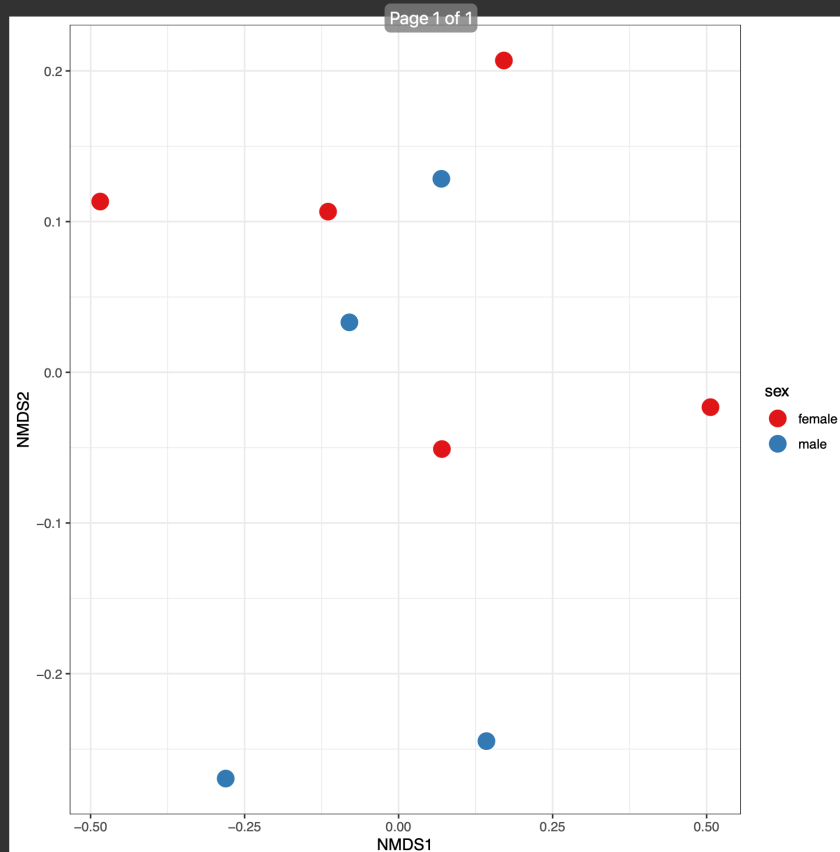
Samples from resident and migratory populations show distinct clustering patterns along the axes. Little to no overlapping

Upload screenshot: NMDS Bray-Curtis plot colored by sex

Choose File



Screensho....46.30 PM



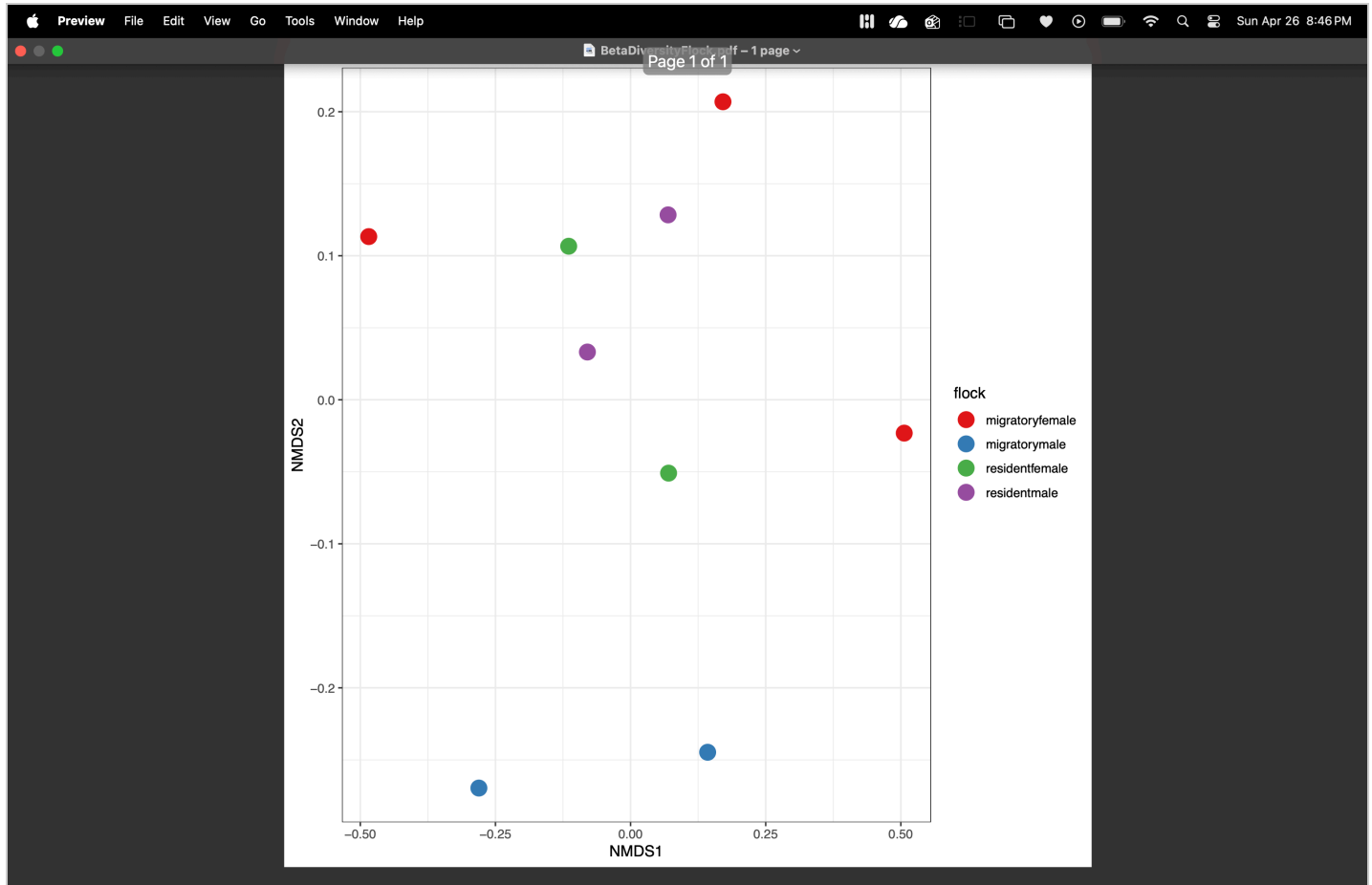
Describe beta diversity patterns by sex.

The distribution of samples by sex does not show clustering. Male and female samples are largely spread out across the plot.

Upload screenshot: NMDS Bray-Curtis plot colored by flock

Choose File

Screensho...46.44 PM



Describe beta diversity patterns by flock.

Grouping by flock shows certain groups, like resident females, cluster more tightly than others.

Part 11 — Taxonomic bar plot

What a taxonomic bar plot shows:

A bar plot summarizes the composition of samples by showing the relative abundance of taxa at a selected taxonomic level.

What taxonomic level did you use?

Phylum

How did you facet the plot?

population

Why was this taxonomic level useful?

Using Phylum is useful because you get all the data needed without much noise. Using Genus or Species creates too much noise and makes the plot difficult to read.

Why was your faceting choice useful?

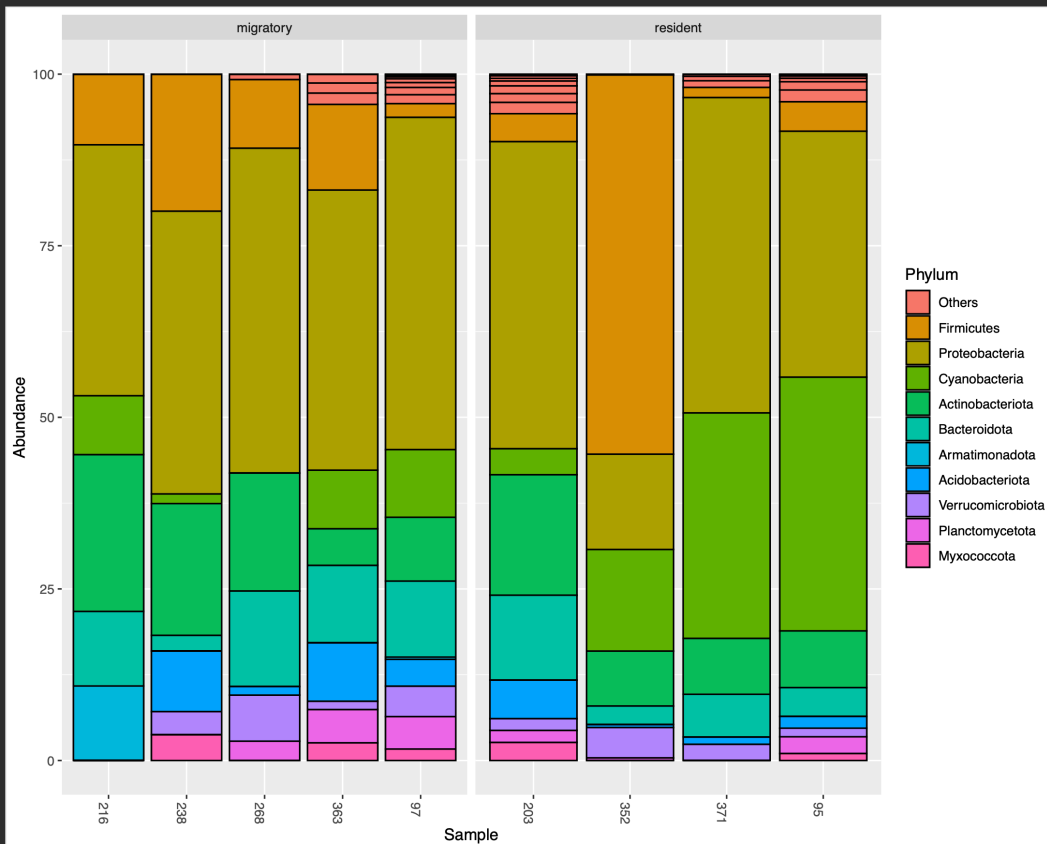
Faceting the data by population is useful because it allows for a comparison of different environmental or behavioral contexts.

Upload screenshot: taxonomic bar plot

Choose File



Screensho...55.21 PM



Which taxa appear to be most abundant?

Across all samples, Proteobacteria and Firmicutes appear to be the most abundant phyla.

Did taxonomic composition differ among groups?

Taxonomic composition does differ among groups. For instance, migratory samples have a higher abundance of Cyanobacteria. Resident samples appear to have a more Actinobacteriota and Bacteroidota.

Part 12 — Overall interpretation

In 3–5 sentences, summarize what your DADA2 results, alpha diversity, beta diversity, and bar plot together suggest about this dataset.

The DADA2 results show and ensure that my data is high quality before running it to make the plots. The microbiome changes depending on whether the birds are migratory or resident. Resident birds have more types of bacteria, while migratory birds are dominated by Proteobacteria and Cyanobacteria. Even though males and females have some small differences, the biggest factor is the bird's population. Overall, the data suggests that a bird's lifestyle and environment shape its microbiome more than its sex.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset

Project/sample set: bird microbiome

Sequencing type: 16s v4 paired-end illumina

Grouping variable: sex and population

Quality profiles

Purpose: The purpose of this step is to read the profile so that we can filter and trim them. This ensures the quality of the reads.

Reverse vs forward quality: The reverse reads are usually lower quality. You would need to truncate more of the plot.

Forward quality drop: The graph doesn't seem to noticeably decline at all.

Reverse quality drop: around 190 bp

Forward truncation: No truncation

Reverse truncation: 190

Why these values: I kept my whole forward quality profile because it doesn't seem to have any lower quality peaks. The reverse plot is always lower quality. I truncated it at 190 bp. I didn't want to but much off because I have a lot of my forward reads.

filterAndTrim

Purpose: Removes low-quality reads and trims poor-quality bases from the ends of sequences.

Parameters used:

Separate filters and trimmers for reverse reads- yes
Truncate reverse at 190

If trimmed too much: There could be some data loss. There is a possibility that we lose too many base pairs. This makes our results incorrect.

If not trimmed enough: This can result in false positives. It will give you data that is untrue.

learnErrors

Purpose: This step removes and corrects the data so we don't misidentify the species.

Model fit: mostly yes

Why important: Learning error rates is important before denoising because it removes any noise without destroying any important data.

dada + mergePairs

Denoising purpose: It removes errors and separates to amplicon data sequences.

ASVs vs OTUs: ASV inference differentiates the sequences by only a single nucleotide. OTU Clustering merges multiple taxa into one taxonomic unit.

mergePairs purpose: The filtered forward and reverse reads are being merged to create a single more accurate sequence.

Need for overlap: The reads need to overlap because it allows for comparison of the two reads and choose the base with the better quality.

One reason reads fail to merge: The amplicon is longer than the forward and reverse reads combined.

Sequence table + sequence length distribution + chimera removal + taxonomy

Sequence table purpose: The table shows the abundance and distribution of the different taxa across the multiple samples.

Rows/columns/cells: Rows are the unique DNA sequence. The columns are the individual sample. The cell values represent the count of the sequence within a specific sample.

Sequence length distribution: The x-axis shows the length of the DNA sequence, and the y-axis shows the count of those sequences. Most of the sequences are approximately 250 to 260 base pairs long. The peak reaches a count around 15,000.

Chimera removal purpose: A chimera is a DNA molecule formed during PCR amplification. It should be removed because it improves accuracy and cleans the data.

Why chimeras are a problem: It can be incorrectly identified as unique taxa, leading to an inaccurate analysis.

Taxonomy purpose: Each ASV is assigned a taxonomic name. It makes you go from having an unknown species to a known species.

Taxonomy limits: There might be some missing information in the data that the database cannot pick up to identify the species.

Alpha diversity

Definition: Alpha diversity describes how diverse a community is.

Observed: Observed measures the richness of a community. The total number of species.

Shannon: Shannon measures the richness and evenness.

Simpson: Simpson measures richness and evenness as well, but it is less sensitive to rare species.

Differences by population: Resident populations have higher observed richness compared to migratory populations. Shannon and Simpson diversity show an overlap between the two groups.

Differences by sex: Observed richness is slightly higher in males than in females. You can also see this in Shannon, where the males have a broader range of diversity values, compared to females.

Differences by flock: The migratory male flock has the highest observed richness and Shannon diversity. The migratory female flock consistently has the lowest diversity across all three.

Beta diversity (NMDS Bray-Curtis)

Definition: Beta diversity measures how similar and different communities are. It compares different communities with each other.

Bray-Curtis: Bray-Curtis measures the differences in the microbiome between samples.

NMDS: NMDS shows the distance between how similar or different the microbiomes are.

Patterns by population: Samples from resident and migratory populations show distinct clustering patterns along the axes. Little to no overlapping

Patterns by sex: The distribution of samples by sex does not show clustering. Male and female samples are largely spread out across the plot.

Patterns by flock: Grouping by flock shows certain groups, like resident females, cluster more tightly than others.

Taxonomic bar plot

Taxonomic level: Phylum

Faceted by: population

Why this taxonomic level? Using Phylum is useful because you get all the data needed without much noise. Using Genus or Species creates too much noise and makes the plot difficult to read.

Why this faceting choice? Faceting the data by population is useful because it allows for a comparison of different environmental or behavioral contexts.

Most abundant taxa: Across all samples, Proteobacteria and Firmicutes appear to be the most abundant phyla.

Did composition differ among groups? Taxonomic composition does differ among groups. For instance, migratory samples have a higher abundance of Cyanobacteria. Resident samples appear to have a more Actinobacteriota and Bacteroidota.

Overall interpretation

The DADA2 results show and ensure that my data is high quality before running it to make the plots. The microbiome changes depending on whether the birds are migratory or resident. Resident birds have more types of bacteria, while migratory birds are dominated by Proteobacteria and Cyanobacteria. Even though males and females have some small differences, the biggest factor is the bird's population. Overall, the data suggests that a bird's lifestyle and environment shape its microbiome more than its sex.